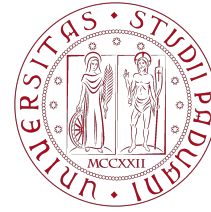


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Tasks 21 & 49

Santarossa, Andrea

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1 | Task 21: Dynamics on network

Task leader(s): *Andrea Santarossa*

Epidemic spreading on static networks: SI, SIS, SIR, SEIR in homogeneous and heterogeneous mean-field approximation.

Modeling epidemic spreading is a challenging problem due to the intrinsic complexity of contagion processes, which emerge from the interplay between biological mechanisms, human behavior, and social interactions. The difficulty of collecting detailed interaction data, together with ethical and privacy constraints, makes it hard to construct fully realistic models. For this reason, epidemic spreading is commonly studied through compartmental models (see Appendix A.1), where individuals are assigned to discrete epidemiological states such as SI, SIS, SIR, and SEIR, allowing for a simplified but effective description of the dynamics. A natural extension of these models is to consider epidemic spreading on contact networks, where nodes represent individuals and links represent possible transmission pathways. This framework makes it possible to investigate how the structure of interactions affects the spreading dynamics. The aim of this task is to analyze SI, SIS, SIR, and SEIR models on static networks within homogeneous and heterogeneous mean-field approximations, comparing different network topologies and studying how the epidemic (invasion) threshold depends on the model parameters and on network heterogeneity.

1.1 | Mathematical framework

The spreading of an epidemic on a network can be described as a stochastic dynamical process in which each node represents an individual and each edge represents a potential transmission channel, as discussed in [4]. The state of each node evolves in time according to the infection and recovery mechanisms, which depend on the states of its neighbors. In general, this leads to a high-dimensional stochastic process that is analytically intractable for large systems. For this reason, mean-field approximations are commonly employed to derive a tractable mathematical description of the dynamics.

In the homogeneous mean-field (HMF) approximation, all nodes are assumed to be statistically equivalent, and the network structure is encoded only through the average degree $\langle k \rangle$. Under this assumption, the probability that a node is infected depends only on the global density of infected individuals. For instance, in the case of a generic SIS-like dynamics, the time evolution of the infected density $I(t)$ can be approximated by:

$$\frac{dI}{dt} = \beta \langle k \rangle I(t) [1 - I(t)] - \mu I(t), \quad (1.1)$$

where β is the infection rate and μ is the recovery rate. This equation captures the competition between infection and recovery processes, assuming that each node interacts

with an average number of neighbors.

A key quantity emerging from this description is the epidemic threshold, which determines whether the infection can invade the system or dies out. This threshold can be expressed in terms of a critical infection rate β_c , defined as:

$$\beta_c = \frac{\mu}{\langle k \rangle}. \quad (1.2)$$

For $\beta < \beta_c$, the infection cannot sustain itself and eventually disappears, while for $\beta > \beta_c$ a finite fraction of the population remains infected in the stationary state.

However, real networks are often heterogeneous, with a broad distribution of node degrees. In this case, the homogeneous approximation fails to capture the role of highly connected nodes. The heterogeneous mean-field approach accounts for this by grouping nodes according to their degree k and introducing degree-dependent densities $I_k(t)$. The evolution equations then become:

$$\frac{dI_k}{dt} = \beta k [1 - I_k(t)] \Theta(t) - \mu I_k(t), \quad (1.3)$$

where $\Theta(t)$ represents the probability that a randomly chosen neighbor is infected. This quantity depends on the degree distribution and introduces a coupling between different degree classes.

Within this framework, the epidemic threshold is again expressed in terms of a critical infection rate, which now explicitly depends on the heterogeneity of the network:

$$\beta_c = \mu \frac{\langle k \rangle}{\langle k^2 \rangle}. \quad (1.4)$$

This result highlights the crucial role of degree fluctuations: in networks with large variance (such as scale-free networks), the second moment $\langle k^2 \rangle$ can diverge, leading to a vanishing epidemic threshold in the thermodynamic limit. As a consequence, even infections with very small transmission rates can spread and persist in highly heterogeneous networks.

1.2 | Epidemic dynamics on homogeneous networks

In this section, we investigate epidemic spreading on a homogeneous network, modeled as an Erdős–Rényi random graph. In this setting, nodes have approximately the same number of connections, and the network can be reasonably described within a homogeneous mean-field approximation.

To validate this assumption, we compare three different descriptions of the dynamics: (i) stochastic simulations on the network, averaged over multiple independent realizations, (ii) the deterministic mean-field ordinary differential equations (ODEs), and (iii) a discrete-time binomial chain approximation [3]. The comparison is shown in Fig. 1.1, where the temporal evolution of the infected fraction $I(t)$ is reported for different epidemic models.

Further details on the implementation of the models and the binomial chain approximation are provided in Appendix A.2.

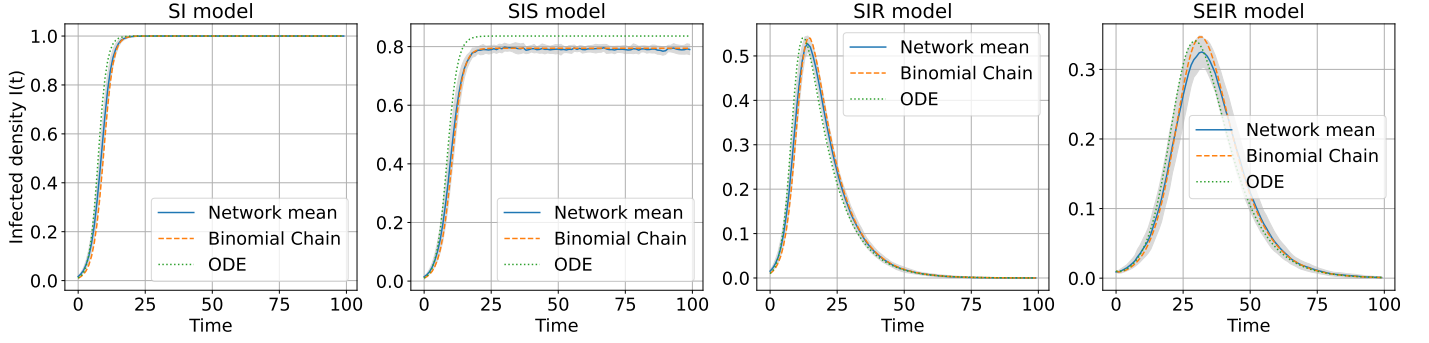


Figure 1.1: Comparison of the infected fraction $I(t)$ for different epidemic models (SI, SIS, SIR, SEIR) on a homogeneous Erdős–Rényi network. The network consists of $N = 1000$ nodes with link probability $p = 0.03$ (average degree $k = 30.5$). The simulations are run for $T = 100$ timesteps with infection rate $\beta = 0.02 > \beta_c$, recovery rate $\mu = 0.1$, and incubation rate $\sigma = 0.2$ (for SEIR). The initial condition is $I_0 = 10$ infected nodes. Results are averaged over 25 independent runs. The shaded area represents one standard deviation around the mean of the network-based simulations, while dashed and dotted lines correspond respectively to the binomial chain approximation and the deterministic ODE model.

1.3 | Epidemic dynamics on heterogeneous networks

We extend the analysis to heterogeneous networks by considering three topologies: Erdős–Rényi, Small-World (Watts–Strogatz), and Scale-Free (Barabási–Albert). For each network, epidemic dynamics (SI, SIS, SIR, SEIR) are simulated, and results are averaged over multiple independent runs. We report the mean fraction of infected nodes over time. Figure 1.2 shows the comparison between the three network topologies, while implementation details are reported in Appendix A.3.

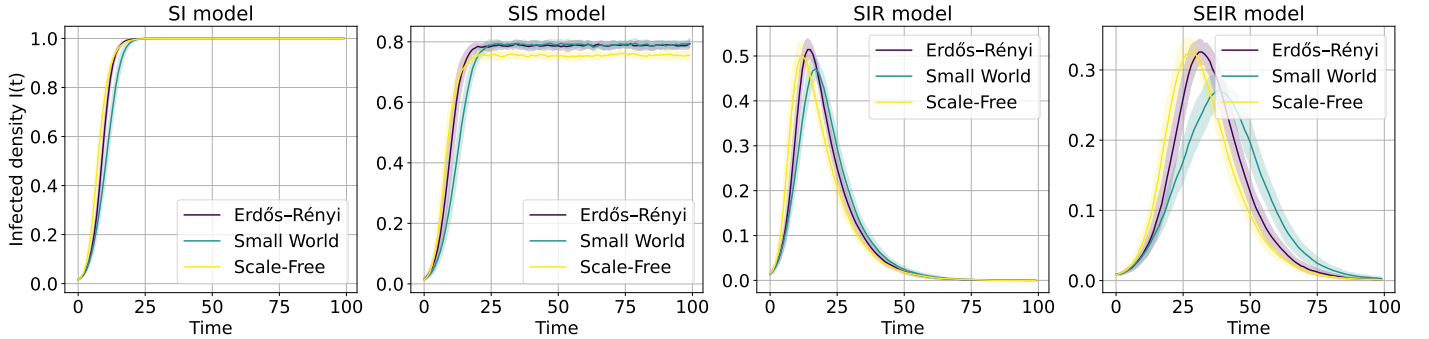


Figure 1.2: Comparison of the infected fraction $I(t)$ for different epidemic models (SI, SIS, SIR, SEIR) across networks with different topological properties: Erdős–Rényi (homogeneous), Small-World (Watts–Strogatz), and Scale-Free (Barabási–Albert). Each panel corresponds to a different epidemic model, while curves represent different network topologies. All networks consist of $N = 1000$ nodes. The Erdős–Rényi network has link probability $p = 0.03$, the Small-World network is generated with $k_0 = 30$ and rewiring probability $p = 0.1$, and the Scale-Free network is constructed with $m = 15$. Simulations are run for $T = 100$ timesteps with infection rate $\beta = 0.02$, recovery rate $\mu = 0.1$, and incubation rate $\sigma = 0.2$ (for SEIR). The initial condition is $I_0 = 10$ infected nodes. Results are averaged over 25 independent runs, and the shaded area represents one standard deviation around the mean.

1.3.1 Epidemic threshold

To investigate the epidemic threshold, we use the Heterogeneous Mean-Field (HMF) approximation introduced in Section 1.1. For each network, we numerically compute the moments of the degree distribution and estimate the corresponding theoretical epidemic threshold β_c , reported in Table 1.1.

For each network, we run 400 simulations of the SIR dynamics for different values

Network	$\langle k \rangle$	$\langle k^2 \rangle$	β_c
Erdős–Rényi	4.02	20.3	0.040
Small-world	4.00	16.8	0.048
Scale-free (BA)	3.99	45.2	0.018

Table 1.1: Degree moments and theoretical epidemic threshold β_c for different network topologies.

of β and compute the final attack rate, defined as $R_\infty = \frac{R(T)}{N}$, averaged over multiple runs, in order to validate the theoretical predictions. The resulting plot is shown in Fig. 1.3, while the details of the network models used and the corresponding degree distributions are reported in Appendix A.3.1.

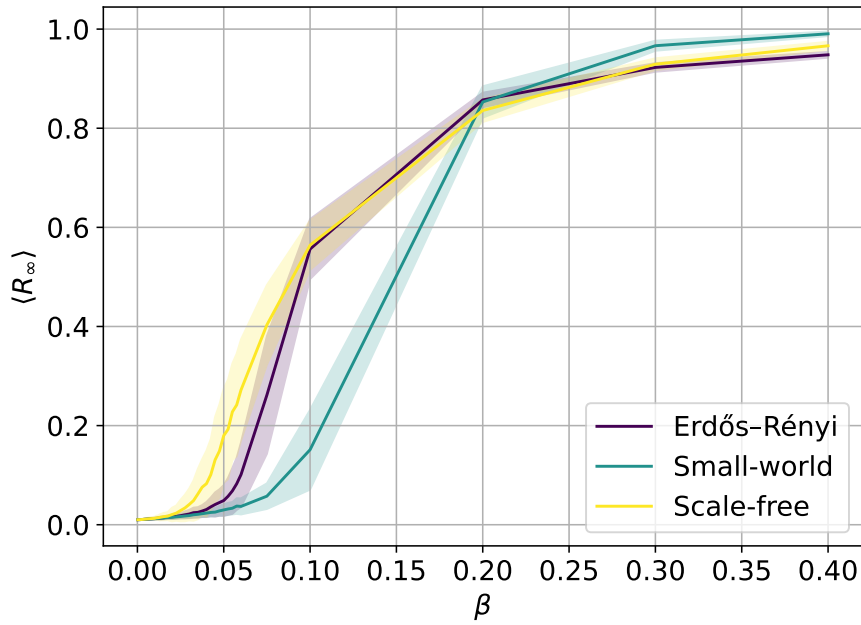


Figure 1.3: Final attack rate $\langle R_\infty \rangle$ as a function of the infection rate β for different network topologies: Erdős–Rényi, small-world, and scale-free (Barabási–Albert). Each point is averaged over $n_{\text{runs}} = 400$ independent SIR simulations with $N = 1000$ nodes and recovery rate $\mu = 0.2$, starting from a single infected node. Shaded regions represent the standard deviation.

1.4 | Conclusion

Real-world contact networks are typically heterogeneous and often display scale-free properties. As shown in this report, scale-free networks are significantly more prone to epidemic spreading due to the reduced (or vanishing) epidemic threshold β_c induced by large degree fluctuations.

This structural property has important implications for control strategies. In particular, random immunization is suboptimal, while more effective approaches exploit network heterogeneity by targeting highly connected nodes. A representative example is acquaintance immunization, which selects individuals through their neighbors rather than uniformly at random (see Appendix A.4), and significantly improves containment efficiency, as shown in the numerical comparison in the Appendix.

2 | Task 49: Temporal winner-loser networks

Task leader(s): *Andrea Santarossa*

Temporal winner-loser networks in WWE, tennis, and football.

This project focuses on the reconstruction of temporal winner-loser networks in WWE/WWF, tennis, and international football, representing competitive interactions as directed weighted graphs aggregated into four-month snapshots. The main WWE/WWF dataset was obtained from Kaggle [1], while ATP and WTA tennis results were collected from the Jeff Sackmann repositories [5, 6]. International football match data were retrieved from the historical results repository by Mart Jürisoo [2]. The aim of the project is to compare the temporal evolution of dominance, centrality, and interaction patterns across sports characterized by different competitive structures through the construction of temporal multilayer networks.

2.1 | Data Preprocessing Pipeline

The preprocessing pipeline follows a similar structure across the three datasets. First, raw match records are imported and cleaned by removing missing or invalid values, while draws are excluded from the football network construction. For the WWE/WWF dataset, only WWE, WWF, WWWF, WCW, ECW, and NXT promotions are retained. In addition, matches labelled as `battle royal`, `royal rumble`, `gauntlet`, `casino`, `bunkhouse`, `elimination chamber`, `rumble`, and `battle royale` are excluded from the winner-loser network, since these match types involve complex all-versus-all elimination dynamics where interpreting the final winner as having directly defeated every participant could artificially inflate the number of winner-loser interactions generated by a single event. Matches are then classified into the interaction layers `singles`, `tag_team`, and `multi_person`.

After the cleaning phase, directed winner-loser edges are created by connecting each winner to the corresponding loser(s), such that an edge $A \rightarrow B$ indicates that player (or team) A defeated player (or team) B . In wrestling, edge construction depends on the match layer, while in tennis and football each match produces a single directed interaction between competitors. Match dates are then converted into temporal snapshots using fixed four-month windows (`January-April`, `May-August`, and `September-December`), allowing interactions occurring within the same period to be aggregated together. Finally, edge weights are summed within each snapshot in order to construct the final temporal weighted networks.

The resulting preprocessed networks are stored in CSV files with a standardized

edge-list representation:

- `wrestling_network.csv`: source, target, weight, snapshot_start, snapshot_end, year, promotion, organization_group, layer.
- `tennis_network.csv`: source, target, weight, snapshot_start, snapshot_end, tour.
- `football_network.csv`: source, target, weight, snapshot_start, snapshot_end, tournament.

Dataset	Interactions	Snapshots	Unique Nodes	Time Span
Wrestling	124961	189	5932	1963–2026
Tennis	1699267	343	65004	1877–2026
Football	36057	400	335	1873–2026

Table 2.1: Summary statistics of the temporal network datasets. *Interactions* indicates the total number of edges, *Snapshots* the number of temporal snapshots, *Unique Nodes* the total number of distinct nodes, and *Time Span* the period between the first and last snapshot.

2.2 | Network analysis

In Appendix B.1 are reported the plots showing the temporal evolution of the network structure. In particular, the evolution of the number of nodes and edges across snapshots is analyzed to characterize long-term structural changes in the three sports systems.

Starting from wrestling, the number of nodes remains relatively stable over time, indicating a consistent pool of active wrestlers across decades, while the number of edges shows an oscillatory behavior with alternating phases of increase and decrease. In tennis, a different pattern emerges, with a clear and persistent increase in the number of edges starting from the 1960s, reflecting the expansion of the professional circuit and the growing number of tournaments. The number of nodes increases only moderately, suggesting that growth is mainly driven by higher match frequency rather than a larger player base. Finally, football exhibits the strongest structural growth, with both nodes and edges increasing significantly over time, especially from the 1960s onward, reflecting the expansion of international competitions and participating national teams.

In Appendix B.2 are reported the plots describing additional structural properties, including density, number of communities, and centrality. Density captures the overall level of interconnectedness of the system, while the number of communities, computed via greedy modularity on the undirected version of the network, reflects its internal organization into competitive clusters. Centrality is measured using PageRank on the reversed directed network, ensuring that importance flows from defeated opponents toward dominant competitors.

The dominance measure is defined as the number of snapshots in which each player (or team) attains the highest PageRank score, providing a notion of sustained superiority over time. In addition, the temporal evolution of rankings is analyzed by tracking

the rank trajectories of the top five players over time. The corresponding results are reported in Figure 2.1, Figure 2.2, and Figure 2.3 for wrestling, tennis, and football respectively. In wrestling, a clear long-term dominance emerges for *Bruno Sammartino* during the 1960s–1980s, while in more recent years dominance appears more fragmented among *Randy Orton*, *John Cena*, and *The Undertaker*. In tennis, the post-2000 era is characterized by a strong rivalry between *Novak Djokovic* and *Roger Federer*, which largely defines the competitive hierarchy of the period. In football, early records (pre-1900) show a clear dominance of *England*, while in the modern era (2000s) a strong superiority of *Brazil* emerges in terms of sustained performance.

Finally, Appendix B.3 reports a more detailed breakdown of dominance and rank evolution across different layers of the wrestling dataset, different circuits in tennis, and different tournaments in football.

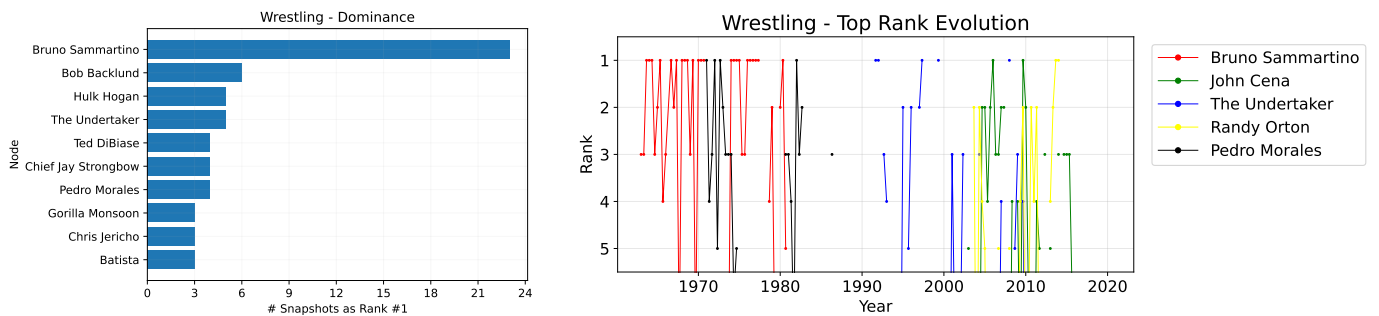


Figure 2.1: Wrestling. Distribution of dominance scores and temporal evolution of ranks over the complete dataset.

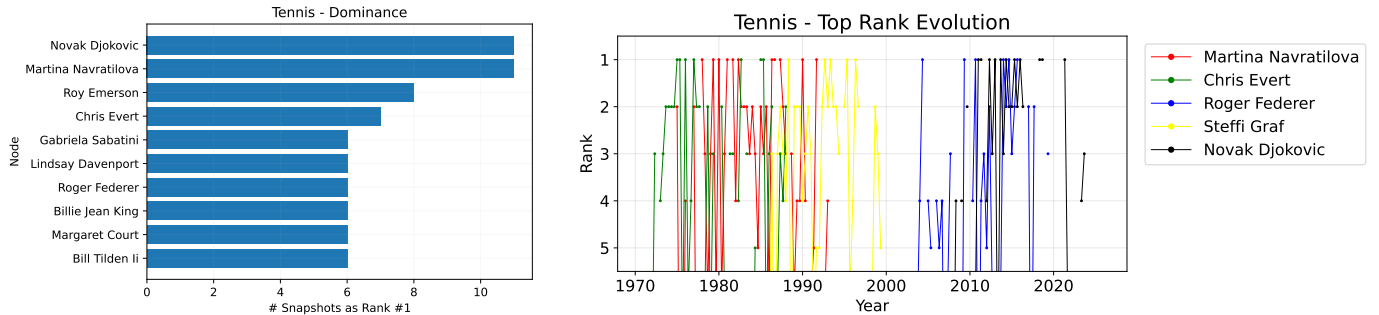


Figure 2.2: Tennis. Distribution of dominance scores and temporal evolution of ranks over the complete dataset.

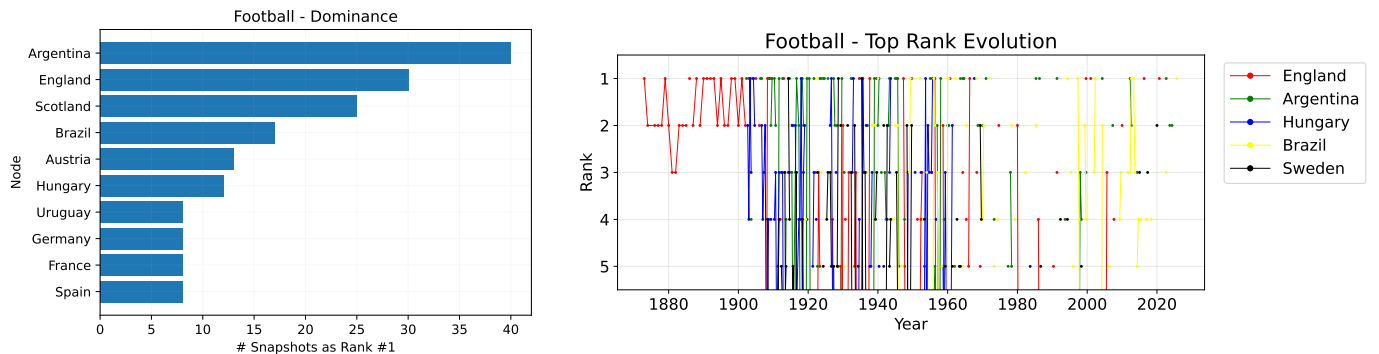


Figure 2.3: Football. Distribution of dominance scores and temporal evolution of ranks over the complete dataset.

3 | Bibliography

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- [5] Jeff Sackmann. Atp tennis match results. https://github.com/JeffSackmann/tennis_atp, 2026. Accessed: 2026-05-25.
- [6] Jeff Sackmann. Wta tennis match results. https://github.com/JeffSackmann/tennis_wta, 2026. Accessed: 2026-05-25.

A | Dynamics on network

A.1 | Compartmental models

We briefly introduce the standard compartmental models used to describe epidemic spreading. In this framework, each individual is assigned to a discrete epidemiological state: susceptible (S), exposed (E), infected (I), or recovered (R). The population is assumed to be well-mixed and normalized such that: $S(t) + E(t) + I(t) + R(t) = 1$.

A.1.1 SI model

The SI model describes a simple contagion process in which susceptible individuals become infected with transmission rate β :

$$\begin{cases} \frac{dS}{dt} = -\beta SI \\ \frac{dI}{dt} = \beta SI \end{cases} \quad (\text{A.1})$$

The stationary points are obtained by imposing $dI/dt = 0$, leading to $I^* = 0$ or $S^* = 0$. The only stable stationary state is the fully infected population $S^* = 0$, $I^* = 1$.

A.1.2 SIS model

In the SIS model, infected individuals recover and return to the susceptible state with rate μ :

$$\begin{cases} \frac{dS}{dt} = -\beta SI + \mu I \\ \frac{dI}{dt} = \beta SI - \mu I \end{cases} \quad (\text{A.2})$$

Using $S = 1 - I$, the stationary points are $I^* = 0$ and $I^* = 1 - \mu/\beta$, where the second solution exists only if $\beta > \mu$, defining the epidemic threshold.

A.1.3 SIR model

In the SIR model, infected individuals recover and acquire immunity:

$$\begin{cases} \frac{dS}{dt} = -\beta SI \\ \frac{dI}{dt} = \beta SI - \mu I \\ \frac{dR}{dt} = \mu I \end{cases} \quad (\text{A.3})$$

The stationary condition implies $I^* = 0$, leading to the disease-free equilibrium $S^* + R^* = 1$.

A.1.4 SEIR model

The SEIR model introduces an exposed state to account for incubation time:

$$\begin{cases} \frac{dS}{dt} = -\beta SI \\ \frac{dE}{dt} = \beta SI - \sigma E \\ \frac{dI}{dt} = \sigma E - \mu I \\ \frac{dR}{dt} = \mu I \end{cases} \quad (\text{A.4})$$

The stationary condition implies $I^* = 0$ and $E^* = 0$, leading to the disease-free equilibrium $S^* + R^* = 1$.

A.2 | Homogeneous networks

This section complements the analysis presented in the main text by providing additional details on the full compartment dynamics and a more in-depth comparison between the different modeling approaches.

The binomial chain approach provides a discrete-time approximation of the spreading process, where the probability that a susceptible node becomes infected is computed assuming independent transmission events from its k neighbors. In particular, the infection probability is given by $1 - (1 - \beta I(t))^k$, which accounts for the cumulative effect of multiple potentially infectious contacts. This formulation can be interpreted as a refined mean-field approximation that partially retains the stochastic nature of the underlying process.

From the comparison, it is evident that the deterministic ODE description is not sufficient to accurately reproduce the dynamics on the network. In particular, for the SIS model, the ODE significantly overestimates the stationary fraction of infected individuals. This discrepancy arises because the ODE assumes a perfectly mixed population and neglects the discrete and probabilistic nature of interactions on the network.

On the other hand, the binomial chain approximation provides a much better agreement with the network simulations, especially for the SI and SIS models. In these cases, the dynamics are primarily governed by local infection and recovery processes, which are well captured by the assumption of independent transmission events.

For the SIR and SEIR models, however, the agreement is less accurate. This is due to the presence of additional compartments and irreversible transitions, which introduce temporal correlations and memory effects that are not fully captured by the

binomial approximation. In particular, the depletion of susceptible individuals and the accumulation of recovered (or exposed) nodes generate correlations that violate the independence assumption underlying the binomial model.

For completeness, the full temporal evolution of all compartments (S, I, R, and E) is reported in Fig. A.1.

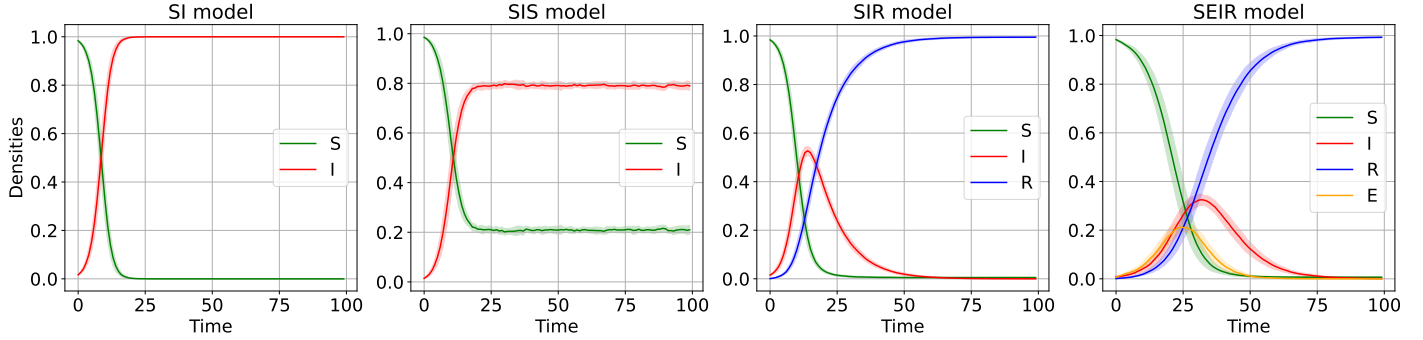


Figure A.1: Temporal evolution of the epidemic compartments (S, I, R, and E when applicable) for different spreading models (SI, SIS, SIR, SEIR) on a homogeneous Erdős–Rényi network. The network consists of $N = 1000$ nodes with link probability $p = 0.03$ (average degree $k = 30.5$). The simulations are run for $T = 100$ timesteps with infection rate $\beta = 0.02$, recovery rate $\mu = 0.1$, and incubation rate $\sigma = 0.2$ (for SEIR). The initial condition is $I_0 = 10$ infected nodes. Results are averaged over 25 independent runs. Solid lines represent the mean-field network averages, while shaded regions correspond to one standard deviation around the mean.

A.3 | Heterogeneous networks

We now extend the analysis to heterogeneous network topologies in order to assess how structural variability in connectivity influences the spreading dynamics and the resulting epidemic behavior.

Across all epidemic models, heterogeneous networks do not only modify the speed of propagation but also the qualitative features of the dynamics. In particular, the Scale-Free network exhibits a systematically faster onset of the epidemic activity and earlier peaks in the transient regimes (SI, SIR, SEIR). This behavior is consistent with the presence of hubs, which dominate transmission pathways and effectively reduce the epidemic threshold.

In contrast, the Small-World network slightly delays the spreading process because of its higher local clustering, which initially constrains diffusion despite the presence of shortcuts.

A.3.1 Epidemic threshold

In Fig. A.2, the degree distributions of the three network topologies used in the main analysis are shown, highlighting their distinct structural properties.

A.4 | Immunization

In real-world complex systems, contact networks often display a highly heterogeneous, scale-free structure, where a small fraction of highly connected nodes (hubs) dominates

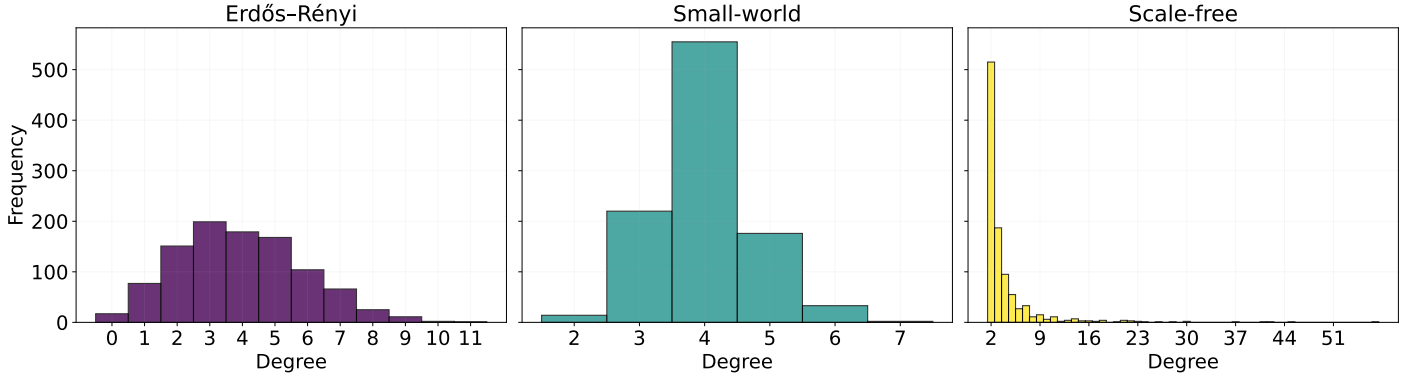


Figure A.2: Degree distributions of the three considered network topologies with $N = 1000$ nodes. The Erdős-Rényi network is generated with connection probability $p = 0.004$. The small-world network follows the Watts-Strogatz model with initial degree $k_0 = 4$ and rewiring probability $p_{\text{rewire}} = 0.2$. The scale-free network is generated using the Barabási-Albert model with attachment parameter $m = 2$.

the spreading dynamics. In this context, a natural and efficient immunization strategy is *acquaintance immunization*. It exploits the fact that, in heterogeneous networks, “*your friends tend to have more friends than you*”. As a result, randomly selecting a neighbor of a randomly chosen node preferentially targets high-degree nodes, without requiring global knowledge of the network.

To investigate the effectiveness of this mechanism, we consider a stochastic SIR model defined on a Barabási-Albert scale-free network. Unlike standard immunization schemes applied at $t = 0$, we introduce a dynamic immunization protocol acting during the epidemic evolution. At each time step, susceptible nodes are progressively removed according to two different strategies: (i) random immunization, and (ii) acquaintance immunization. As shown in Fig. A.3, the effectiveness of acquaintance immunization strongly depends on the epidemic time scale. For slow spreading regimes (small β), the acquaintance strategy significantly reduces the epidemic peak and delays the outbreak compared to random immunization. In contrast, for fast spreading regimes (large β), the difference between the two strategies becomes negligible, as the epidemic evolves too rapidly for the network structure to be effectively exploited by the immunization protocol.

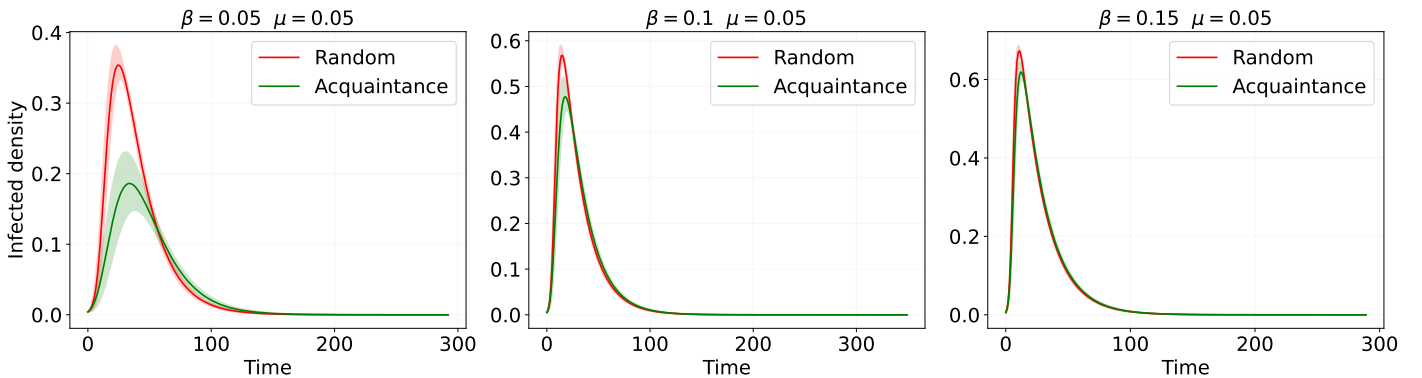


Figure A.3: SIR dynamics with random and acquaintance immunization on a Barabási-Albert scale-free network ($N = 3000$, $m = 3$). The recovery rate is $\mu = 0.05$, with infection rates $\beta \in \{0.05, 0.10, 0.15\}$. A fraction $\alpha = 0.01$ of susceptible nodes is immunized at each time step. Each simulation starts from $I_0 = 10$ randomly selected infected nodes. Results are averaged over 400 realizations. Solid lines show the mean infected density, shaded areas one standard deviation.

B | Temporal winner-loser networks

B.1 | Network size evolution

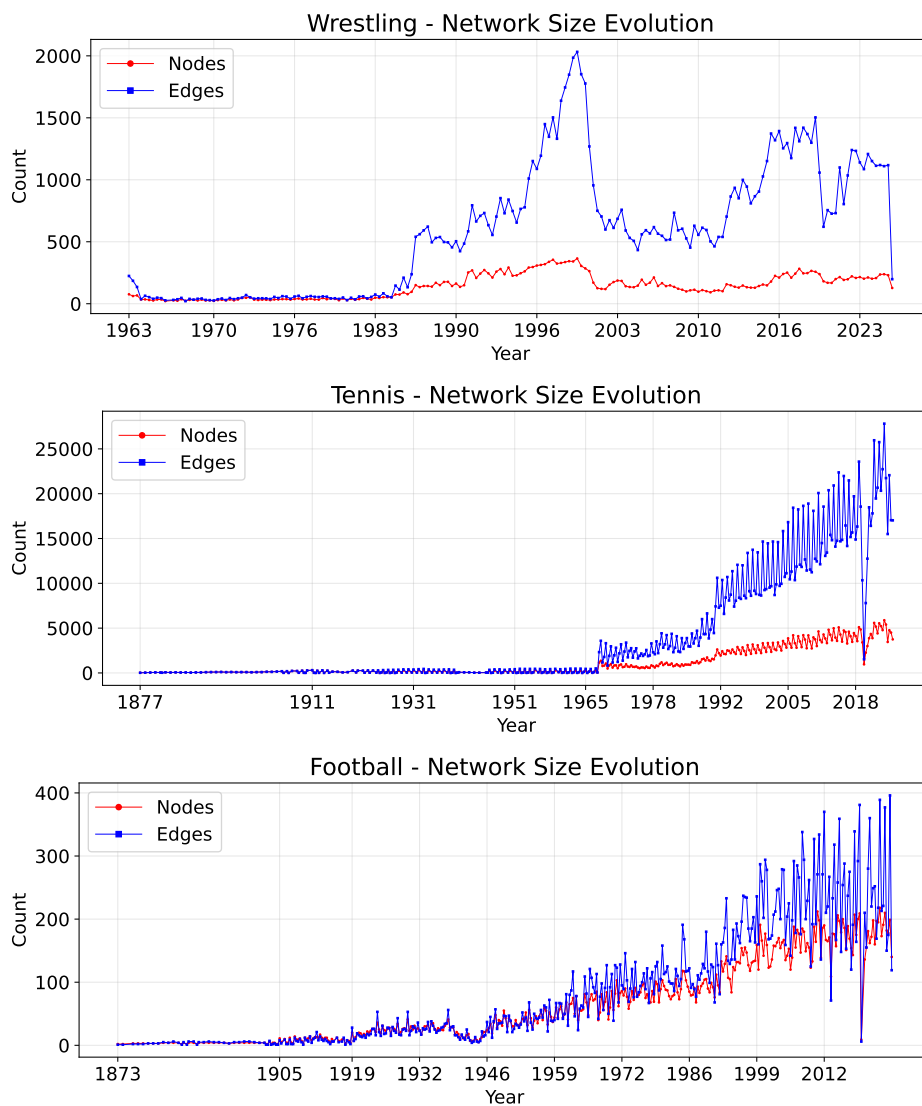


Figure B.1: Evolution of the number of nodes and edges across temporal snapshots for the Wrestling, Tennis, and Football datasets.

B.2 | Temporal evolution of structural network properties

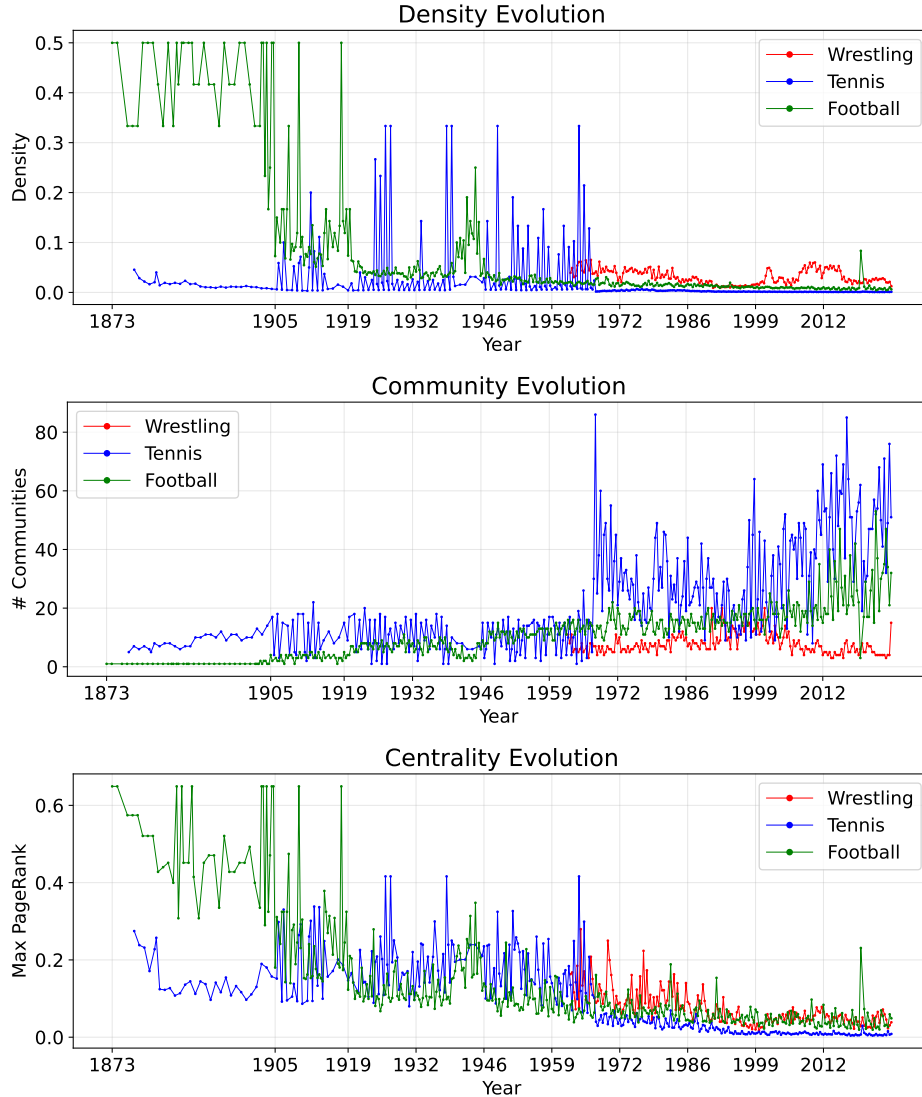


Figure B.2: Temporal evolution comparison of density, number of communities, and centrality across Wrestling, Tennis, and Football datasets.

Football shows the highest density in the early periods, followed by a strong long-term decline as the network expands and becomes sparser. Tennis exhibits a more irregular pattern, with intermittent peaks in density, especially during periods of rapid tour expansion. Wrestling remains comparatively stable and low in density, reflecting a more constrained and consistent interaction structure. Regarding community structure, tennis displays the strongest increase over time, indicating a progressively more fragmented system. Football also shows a gradual growth in the number of communities, while wrestling remains relatively stable with fewer clusters. Finally, centrality (max PageRank) is highest in early football and tennis periods, but decreases over time, suggesting a progressive decentralization of dominance. Wrestling maintains lower and more stable centrality values, consistent with a more evenly distributed competitive hierarchy.

B.3 | Dominance and rank evolution analysis

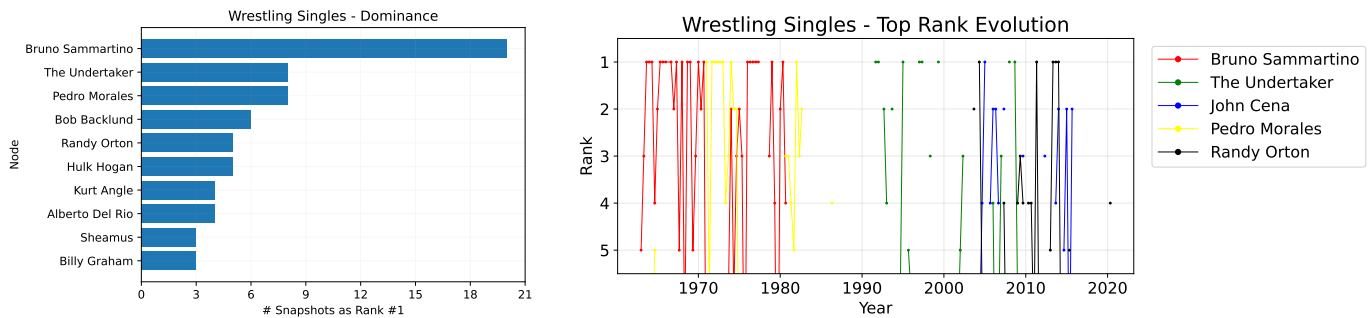


Figure B.3: Wrestling Singles layer. Dominance score distribution (left) and temporal rank evolution (right).

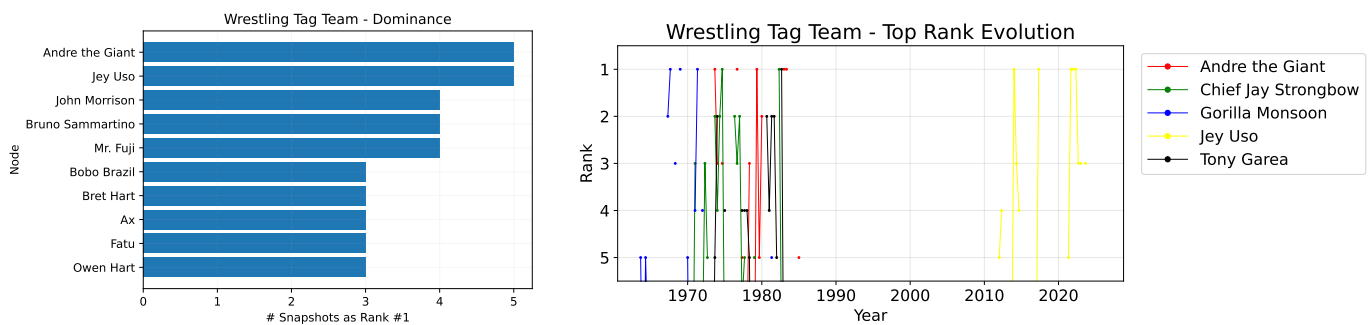


Figure B.4: Wrestling Tag Team layer. Dominance score distribution (left) and temporal rank evolution (right).

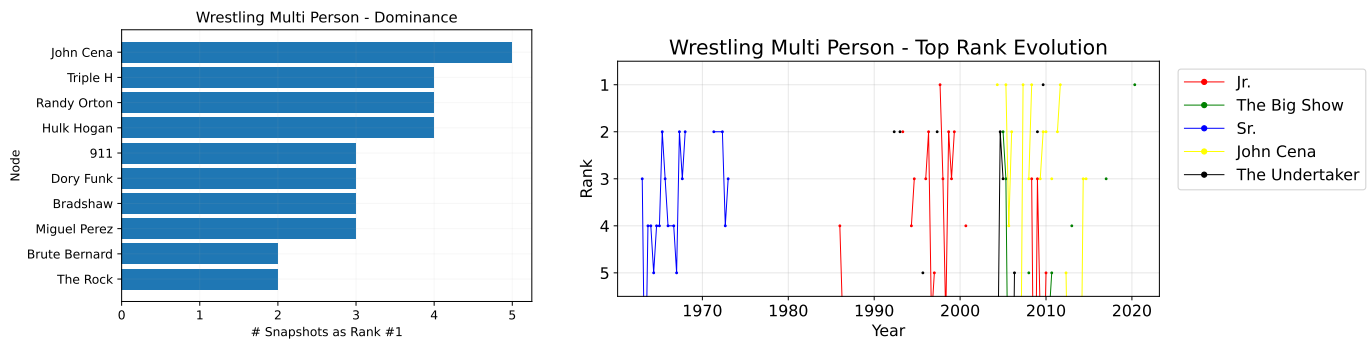


Figure B.5: Wrestling Multi Person layer. Dominance score distribution (left) and temporal rank evolution (right).

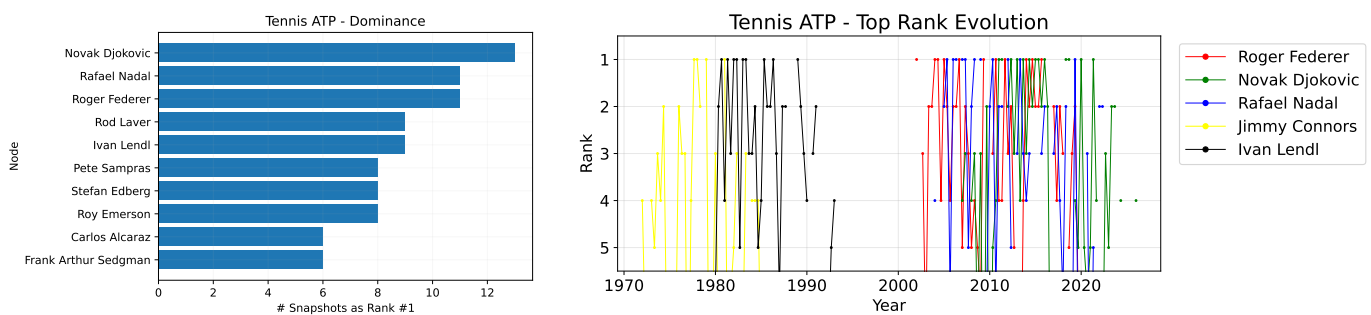


Figure B.6: ATP tennis circuit. Dominance score distribution (left) and temporal rank evolution (right).

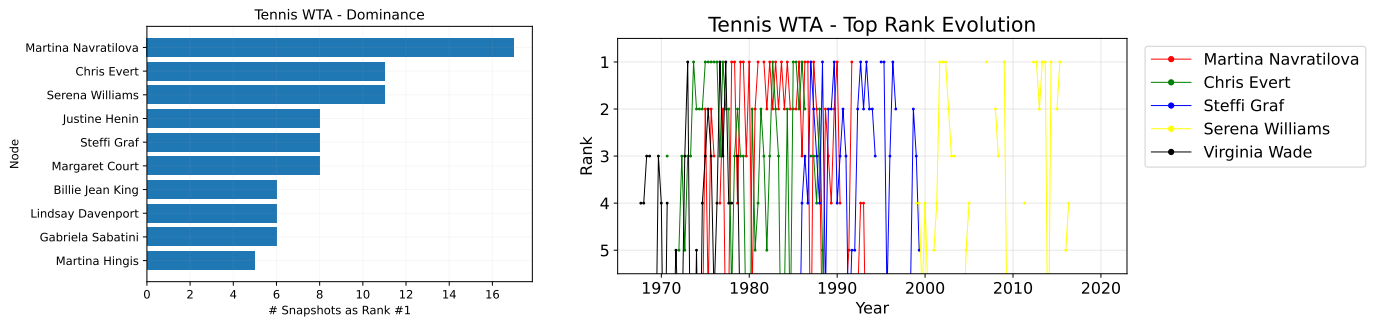


Figure B.7: WTA tennis circuit. Dominance score distribution (left) and temporal rank evolution (right).

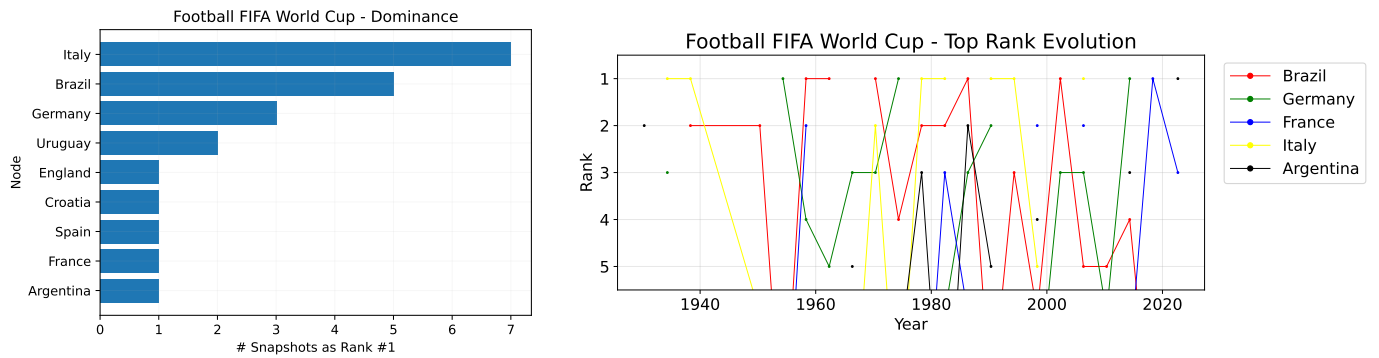


Figure B.8: FIFA World Cup. Dominance score distribution (left) and temporal rank evolution (right).

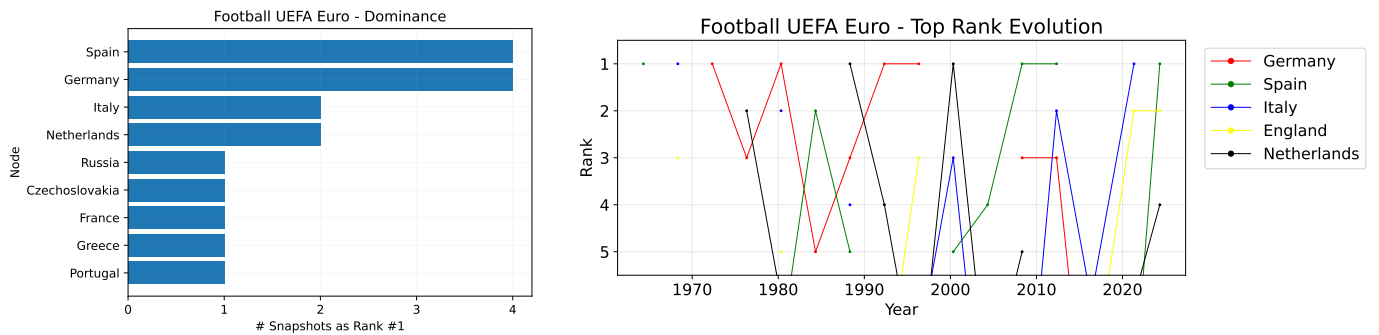


Figure B.9: UEFA European Championship. Dominance score distribution (left) and temporal rank evolution (right).

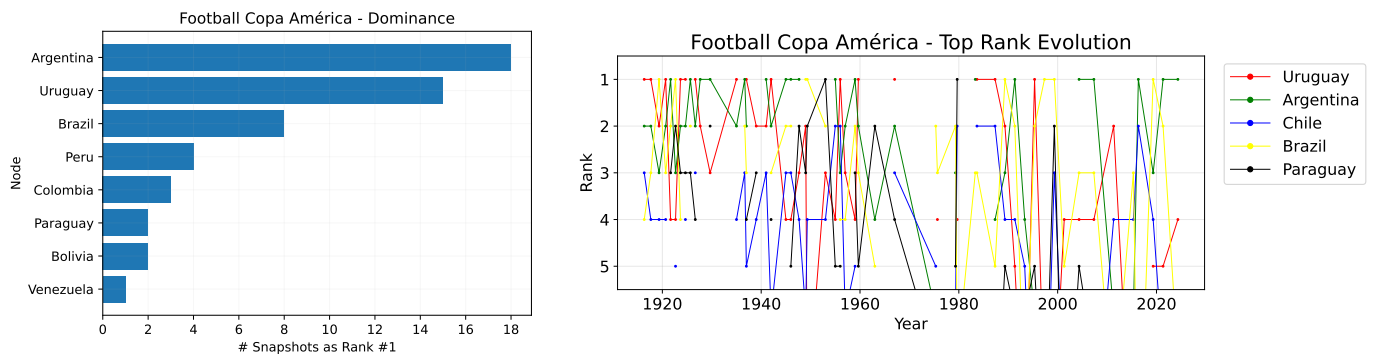


Figure B.10: Copa América. Dominance score distribution (left) and temporal rank evolution (right).